[23] The Global Landscape of Neural Networks: An Overview

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RQ1 / RQ4 — review of landscape results, not a new all-networks landscape theorem.

Problem. The non-convexity of network training does not by itself tell us whether local optimization finds poor solutions. Which models have favorable global landscapes? **Method.** The authors survey analytical results for structured neural-network models and separate favorable geometry from cases with spurious minima. **Result.** A conditional landscape picture, not a universal statement that neural networks have no bad local minima.

1. Three distinct objects

Given training data and parameters θ, let $f(\theta) = \frac{1}{n}\sum_{i=1}^{n} \ell(y_i, \widehat{y}_i(\theta))$. We must distinguish

- **Critical point:** $\nabla f(\theta) = 0$ (when differentiable).
- **Local minimum:** $f(\theta) \le f(\theta + u)$ for all sufficiently small u.
- **Global minimum:** $f(\theta) \le f(\vartheta)$ for every feasible ϑ.

Even a theorem about *all local minima* does not automatically characterize all critical points, and even zero training error does not imply test-set generalization.

2. Why factorized linear models are informative

Consider $f(A, B) = \frac{1}{2}\|AB - M\|_F^2$. This is non-convex in (A, B) because the product is bilinear, while the corresponding objective $F(Z) = \frac{1}{2}\|Z - M\|_F^2$ is convex in Z. Matrix factorization can create saddles and symmetries without necessarily creating spurious local minima in suitable settings.

Toy calculation (illustrative). $f(a, b) = \frac{1}{2}(ab - 1)^2$ has $(0, 0)$ as a critical point. Its Hessian at the origin is $\left(\begin{smallmatrix} 0 & -1 \\ -1 & 0 \end{smallmatrix}\right)$, with eigenvalues $-1, +1$: this is a strict saddle, not a bad local minimum. A nonlinear-activation result cannot be deduced from this linear example.

3. Width, activation, data and loss each matter

Landscape results for deep/wide nonlinear networks commonly impose activation properties, sample-separation conditions, sufficiently large hidden-layer widths and/or non-degenerate weights. **Positive results** may show that certain local minima are global or that almost all local minima are global. **Negative examples** demonstrate genuine spurious minima even for simple ReLU networks. Neither family of results overrides the hypotheses of the other.

The general second-order test at a critical point uses $H = \nabla^2 f$. A negative eigenvalue implies a descent direction; $H \succeq 0$ does *not* rule out higher-order saddles. For example,

$$g(x, y) = x^2 - y^4, \quad \nabla g(0, 0) = 0, \quad \nabla^2 g(0, 0) = \mathrm{diag}(2, 0),$$

although $g(0, y) < g(0, 0)$ for small $y \neq 0$.

4. What the review contributes — and what it does not

Attribution. This is a review of earlier theorem families, including linear, overparameterized and restricted nonlinear models. It does not prove that general practical ReLU, convolutional or transformer training landscapes have no spurious minima; the assumptions are highly model dependent.

Survey connection. RQ1: organizes minima and saddle geometry; RQ4: makes precise why theorem assumptions matter. **Our Notes.** The right question is not "is a neural-network loss non-convex?" but "which added structural conditions make that non-convexity benign?" The bad-minimum question remains architecture- and data-dependent.

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